

Assignments Continuous Integration Using Jenkins

Submitted to - Vikul Sir

Date of Submission -29/MAR/2026

Submitted by – NALATHATI RAHUL

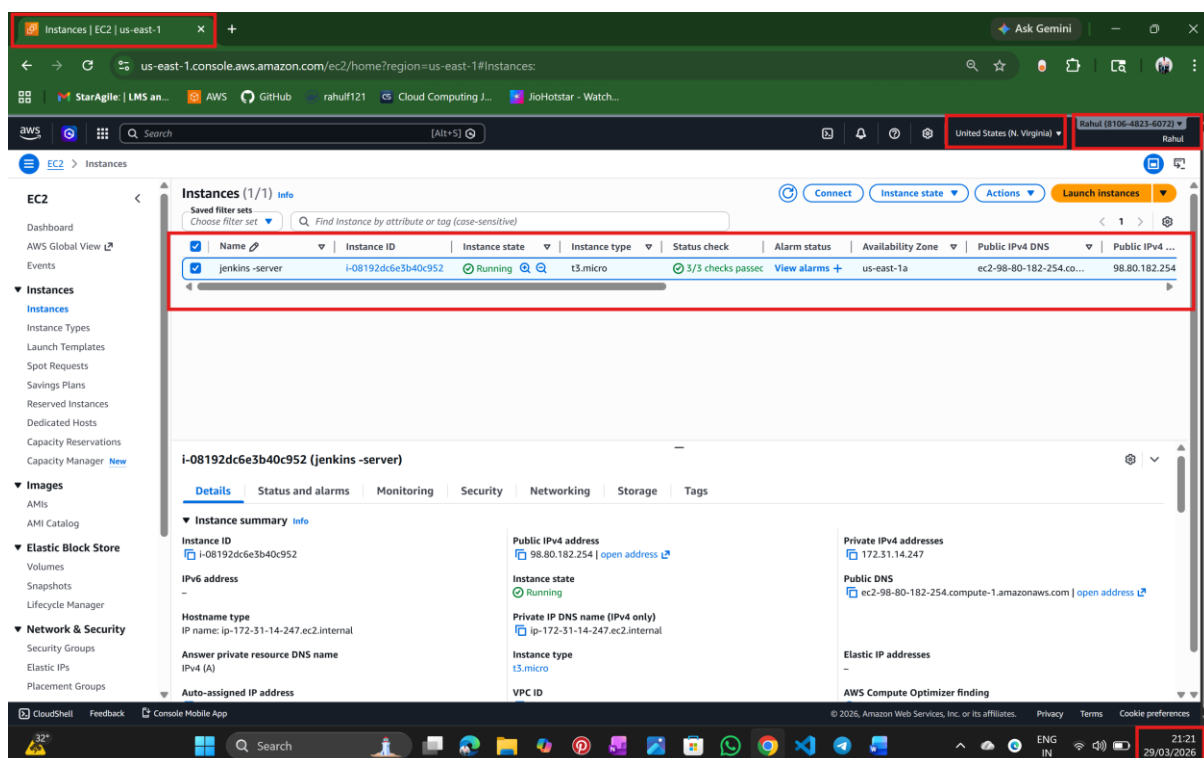
L4 - Automate the CI/CD Pipeline using GitHub webhook and Poll-SCM

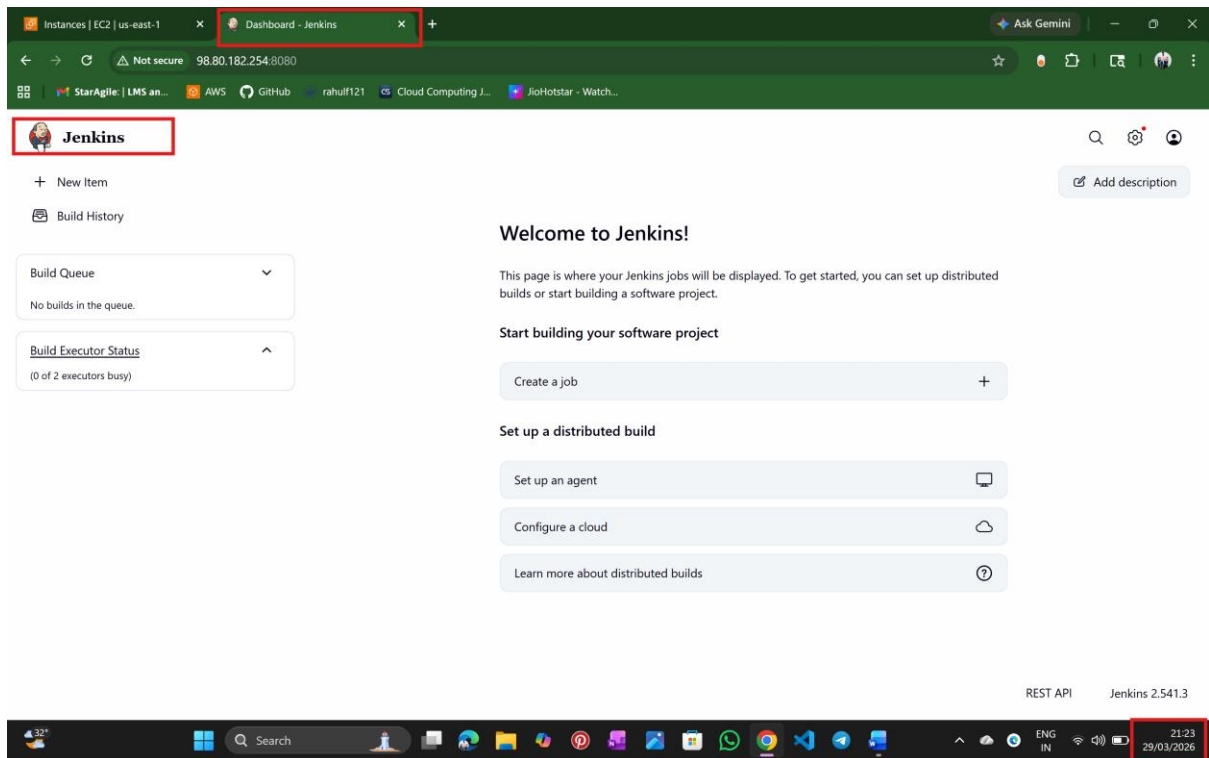
Step 1: Install Jenkins

→ Install Jenkins on your system or EC2 instance

→ Start Jenkins service

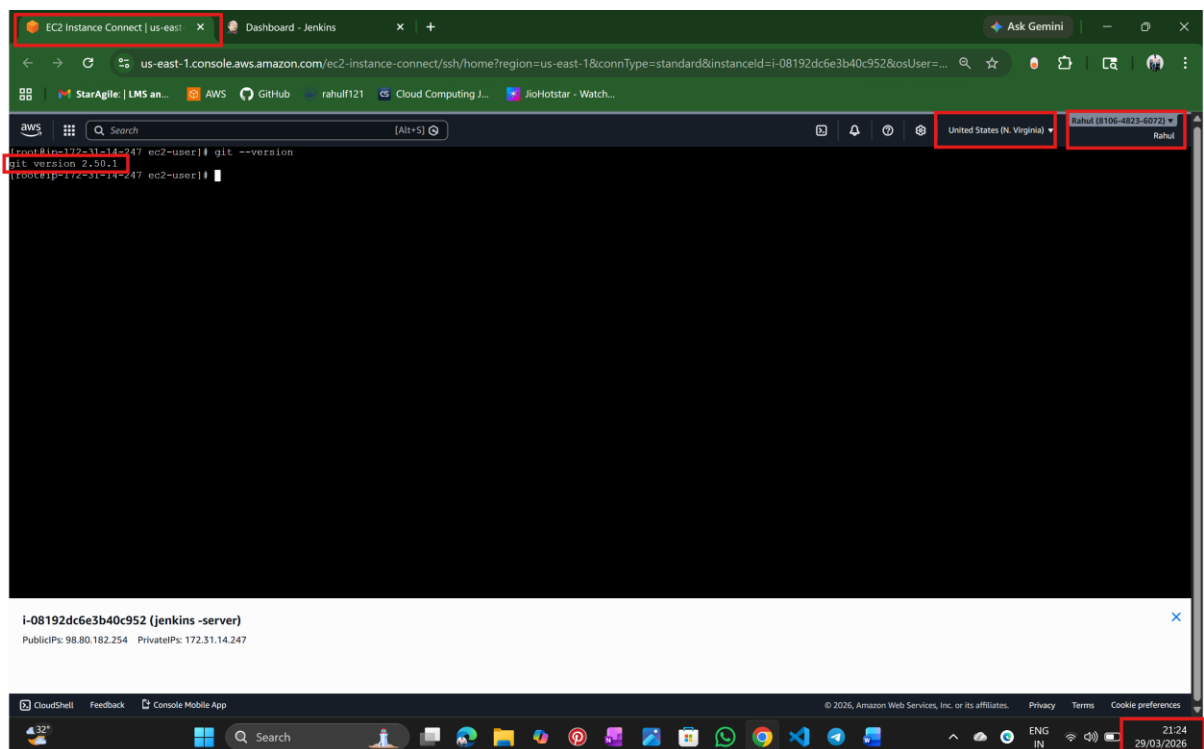
→ Open Jenkins in browser:





Step 2: Install Required Tools

→Git

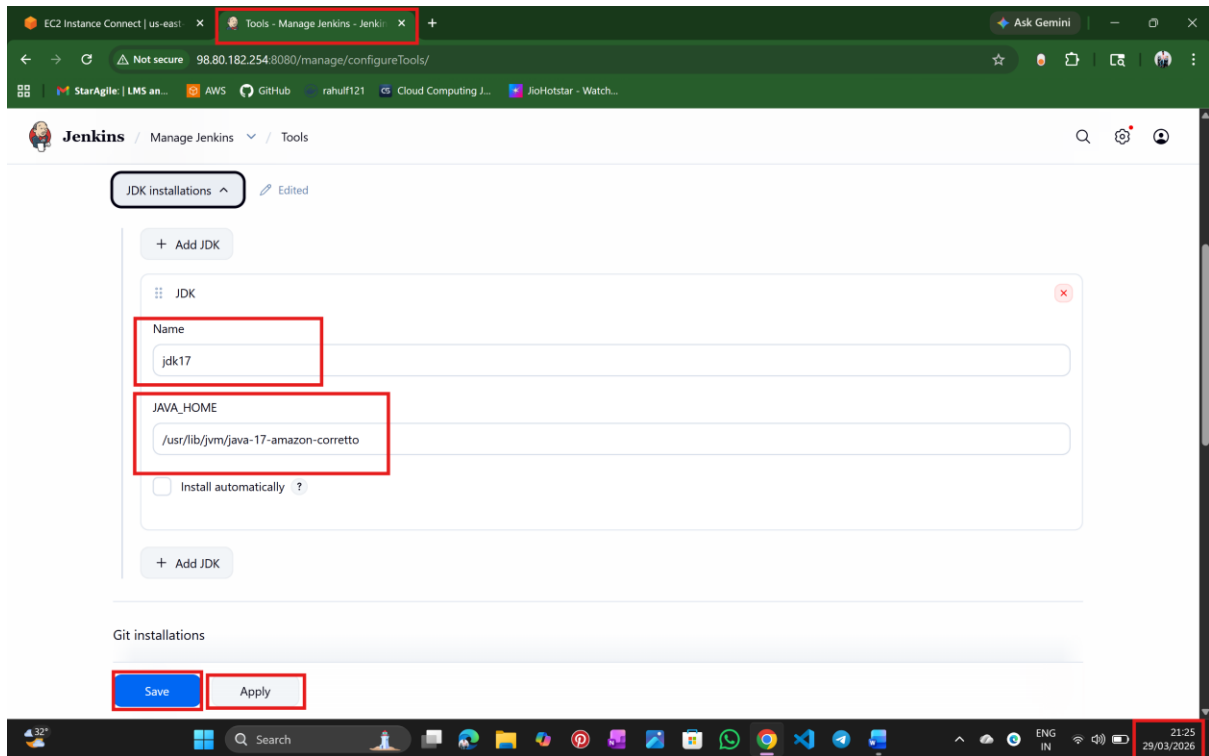


Step 3: Configure Global Tools in Jenkins

Add:

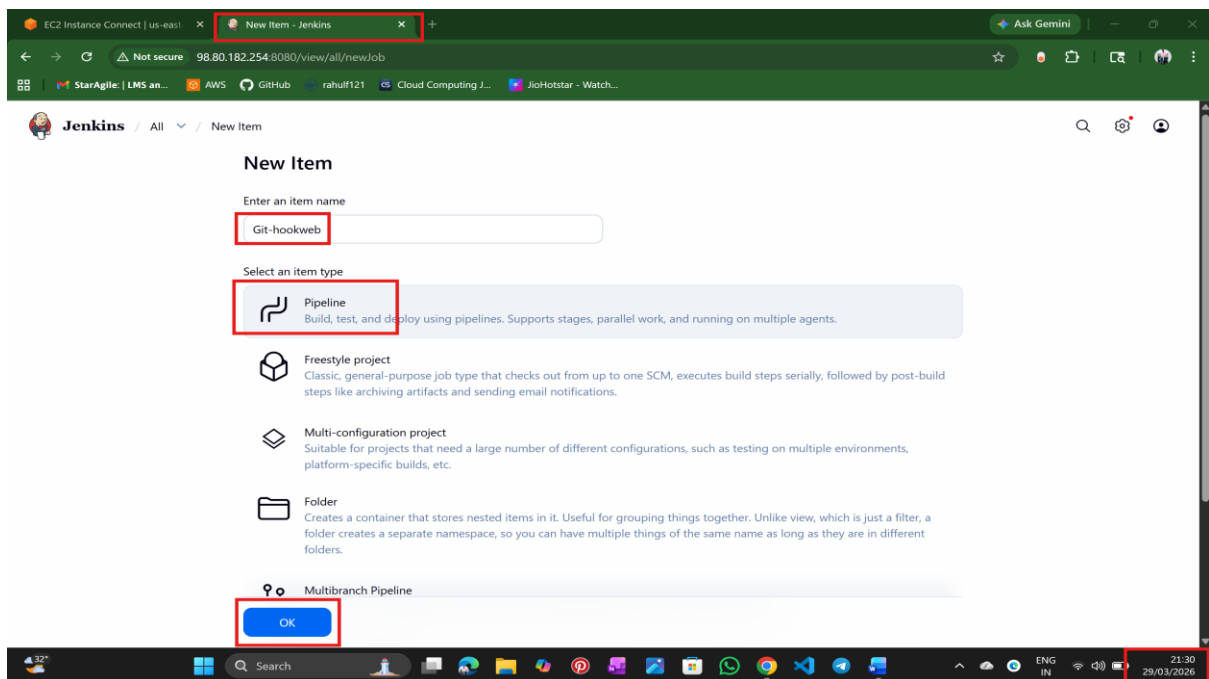
→JDK

→Maven

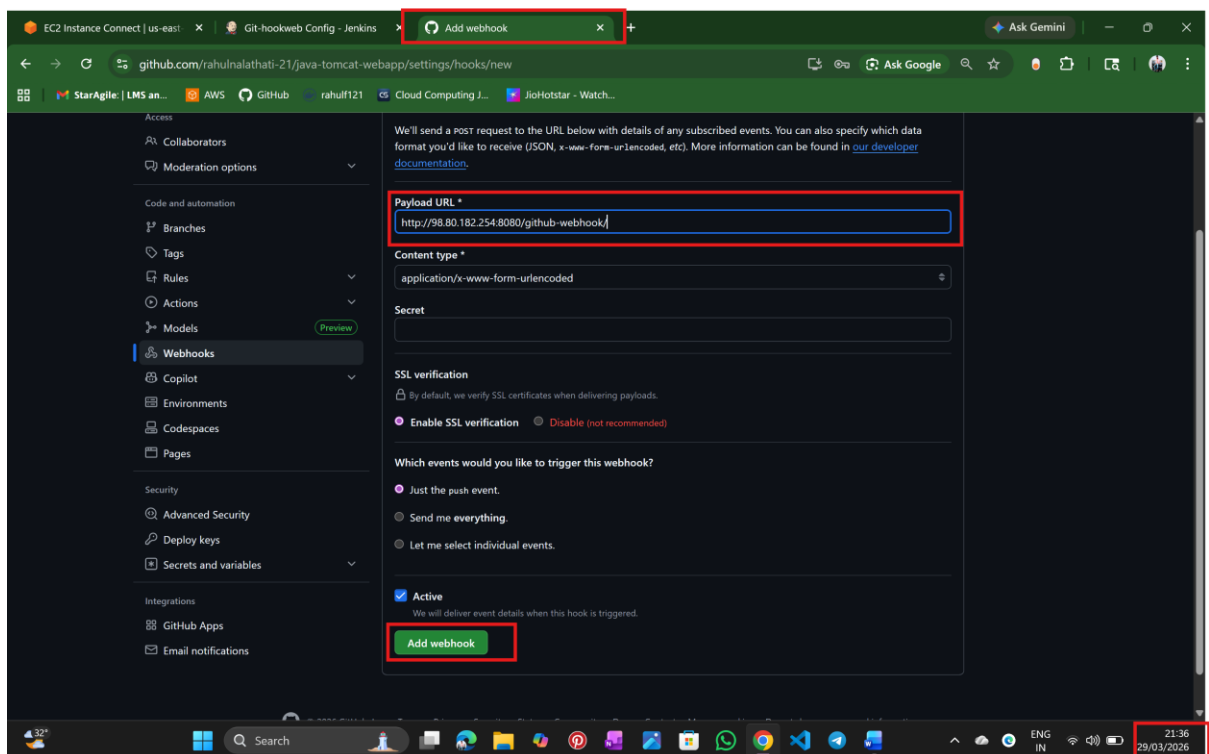
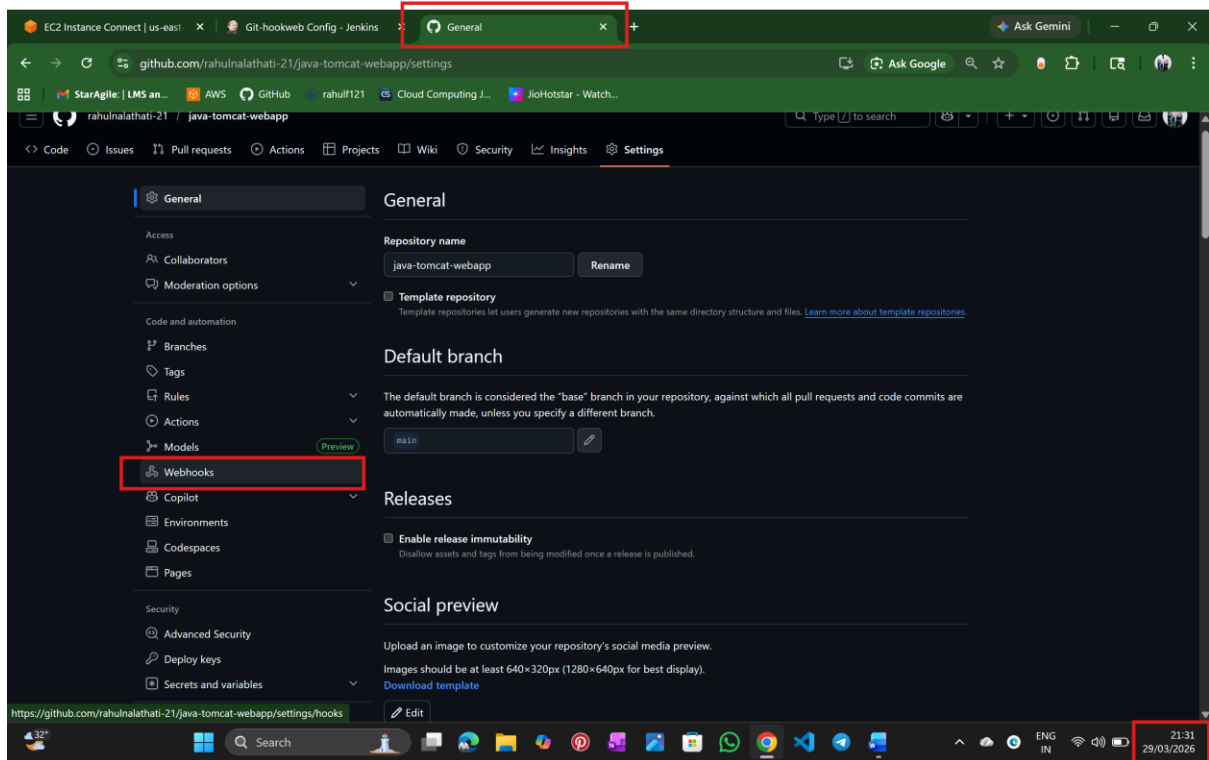


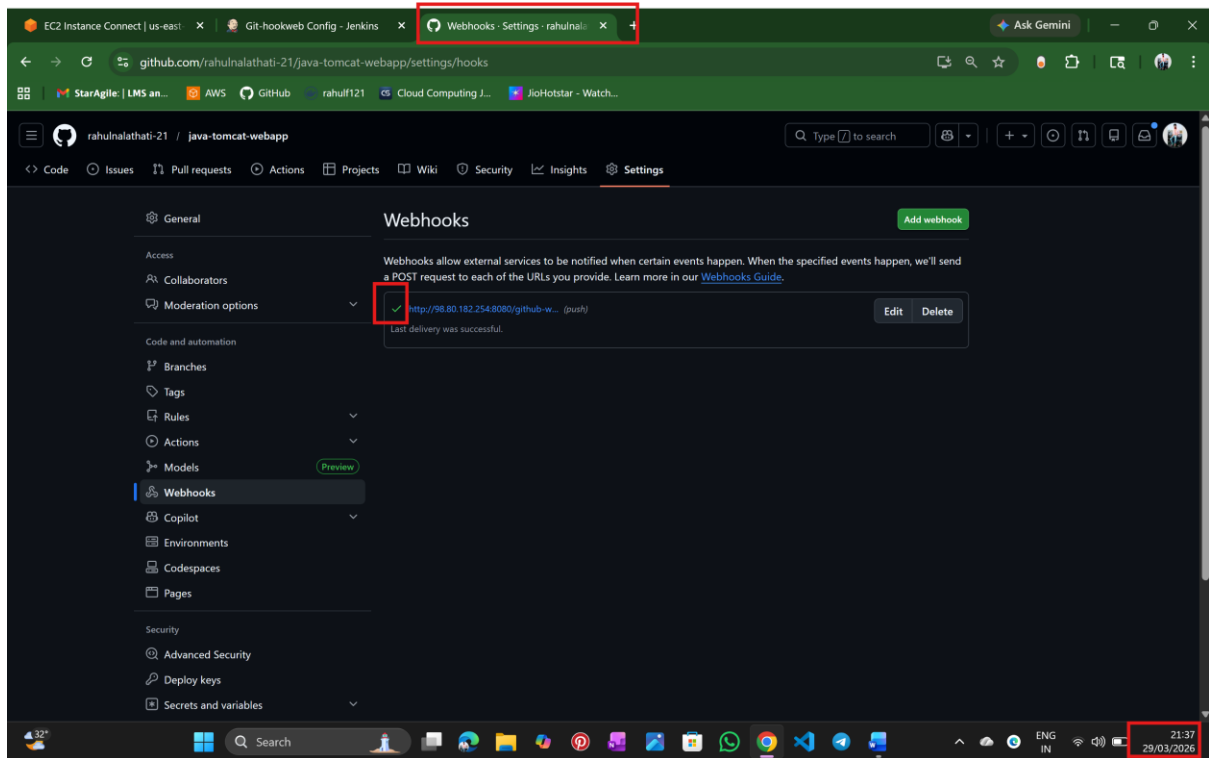
Step 4: Create Jenkins Pipeline Job

→Click New Item →Enter job name: Git-hookweb →Select Pipeline →Click OK



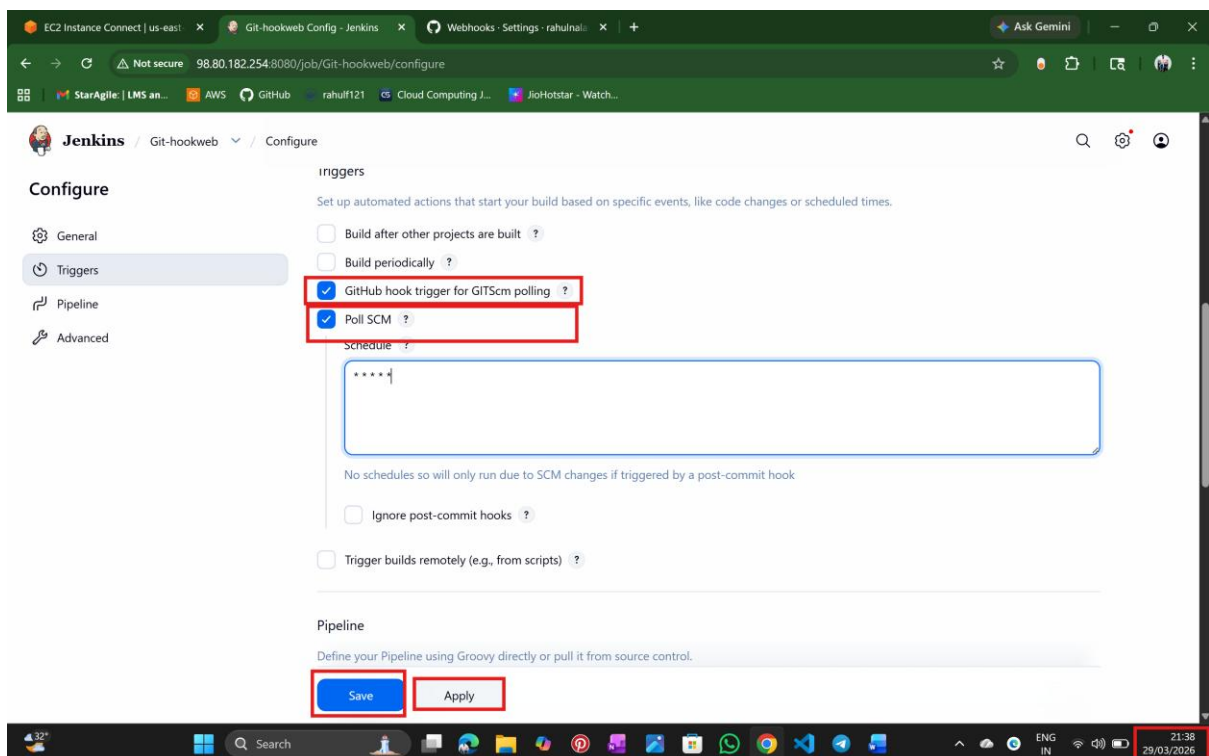
Step 5: add webhook in git hub





Step 6: Confirm GitHub hook

→enable poll SCM



Step 6: pipeline script

```
pipeline {
    agent any

    stages {
        stage('git checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/rahuInalathati-21/java-tomcat-webapp.git'
            }
        }
        stage('maven build') {
            steps {
                sh 'mvn clean package'
            }
        }
        stage('Artifact to local') {
            steps {
                archiveArtifacts artifacts: '**/*.war', fingerprint: true
            }
        }
        stage('Deploy to tomcat') {
            steps {
                deploy adapters: [tomcat9(alternativeDeploymentContext: '', credentialsId: 'tomcat-creds', path: '',
url: 'http://100.48.210.151:8080')], contextPath: 'java-App', war: '**/*.war'
            }
        }
    }
}
```

Step 7: launch Tomcat server

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Instances' page is active, displaying a table of EC2 instances. The 'Tomcat-server' instance is highlighted with a red box. Below the table, the details for the 'Tomcat-server' instance (ID: i-084c69e622d213868) are shown, including its state (Running), instance type (t3.micro), and public IP address (100.48.210.15).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4
Tomcat-server	i-084c69e622d213868	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-100-48-210-151.co...	100.48.210.15
jenkins-server	i-08192dc6e3b40c952	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-98-80-182-254.co...	98.80.182.254

i-084c69e622d213868 (Tomcat-server)

Instance summary

Instance ID: i-084c69e622d213868

Public IPv4 address: 100.48.210.15 | [open address](#)

Instance state: Running

Private IP4 addresses: 172.31.1.2

Public DNS: ec2-100-48-210-151.compute-1.amazonaws.com | [open address](#)

Private IP4 address: 172.31.1.2

Public DNS: ec2-100-48-210-151.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses: -

AWS Compute Optimizer finding: -

Step 8: Login into tomcat server

The screenshot shows the Tomcat Web Application Manager interface. The title 'Tomcat Web Application Manager' is highlighted with a red box. The interface displays a list of applications and a deployment section.

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

Deploy directory or WAR file located on server

Step 9: Therefore the , web application successfully deployed

The screenshot shows the Jenkins web interface for a pipeline named 'Git-hookweb'. The 'Stage View' tab is active, displaying a progress bar for the current build. The stages and their durations are:

Stage	Duration
git checkout	795ms
maven build	4s
Artifact to local	184ms
Deploy to tomcat	541ms

The 'Average stage times' are also shown: (full run time: ~7s). The 'Permalinks' section lists the last build (#1) as completed 13 minutes ago. The 'Builds' list on the left shows two builds: #2 (4:34 PM) and #1 (4:21 PM). The system clock at the bottom right indicates 22:05 on 29/03/2026.

The screenshot shows the 'DevOps Deployment' page. A central message box states:

CI/CD Pipeline Success!

The Java Application was built by Jenkins and is now running on Tomcat.

Deployment Time: Sun Mar 29 16:35:54 UTC 2026

Project Name: devops-project

The system clock at the bottom right indicates 22:06 on 29/03/2026.

-----END-----

NALATHATI RAHUL

